1. Pareto Distribution

The Pareto distribution with some shape parameter $\alpha > 0$ has CDF and PDF

$$F(x) = 1 - \left(\frac{1}{x+1}\right)^{\alpha} \quad \text{and} \quad f(x) = \frac{\alpha}{(x+1)^{\alpha+1}}$$

for $x \geq 0$. Use the probability integral transform to find a transformation that takes $U \sim \text{Uniform}\,[0,1]$ to $X \sim \text{Pareto}(\alpha)$.

write $U = F(x)$ and invert.

$$U = 1 - \left(\frac{1}{x+1}\right)^{\alpha}$$

$$\left(\frac{1}{x+1}\right)^{\alpha} = 1 - U$$

$$x = (1-U)^{-1/\alpha} - 1 \longrightarrow \underline{\text{extra}} \quad U^{1/\alpha} - 1$$

because if $U \in \text{Uniform}\,[0,1]$
then so is $1 - U \in \text{Uniform}\,[0,1]$.

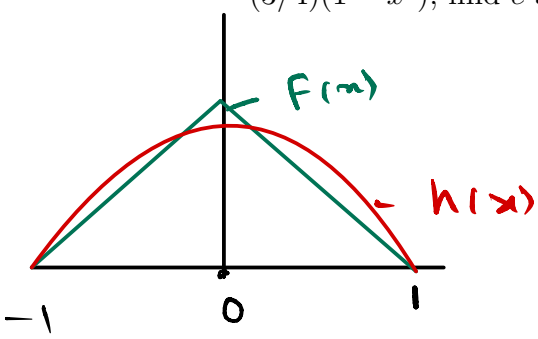
1. Pareto Distribution

2. Triangular Distribution

The triangular distribution has CDF and PDF

$$F(x) = \begin{cases} (1+x)^2/2, & x \in [-1,0] \\ 1-(1-x)^2/2, & x \in [0,1] \end{cases} \quad \text{and} \quad f(x) = \begin{cases} 1+x, & x \in [-1,0] \\ 1-x, & x \in [0,1] \end{cases}$$

for $x \in [-1,1]$. The acceptance-rejection methods requires us to write $f(x) = cg(x)h(x)$. If we use acceptance-rejection with the the spherical law, $h(x) = (3/4)(1-x^2)$, find c and $g(x)$ and the probability of an acceptance. $\to 1/c$



Step!

$$c = \max_{x \in [-1,1]} \frac{f(x)}{h(x)}$$

$(1-x)(1+x)$

$$= \max_{x \in [-1,1]} \begin{cases} 1+x \,/\, 3/4\,(1-x^2) & x \in [-1,0] \\ 1-x \,/\, 3/4\,(1-x^2) & x \in [0,1]. \end{cases}$$

$$= \frac{4}{3} \max \begin{cases} \frac{1}{1-x}, & x \in [-1,0], \text{ max at } x=0 \\ \frac{1}{1+x}, & x \in [0,1], \text{ max at } x=0 \end{cases}$$

$$\text{Prob(accept)} = \frac{1}{c} = 3/4$$

$$g(x) = \frac{1}{c} = \frac{f}{h} = \begin{cases} \frac{1}{1-x}, & x \in [-1,0] \\ \frac{1}{1+x}, & x \in [0,1] \end{cases}$$

2. Triangular Distribution

3. Beta Distribution

The Beta$(3, 1)$ distribution has CDF and PDF

$$F(x) = x^3 \quad \text{and} \quad f(x) = 3x^2$$

for $x \in [0, 1]$. For the ratio of uniforms method, we choose $r(z) = z^2$ for $z \in [0, 1]$ and $r(z) = 0$ otherwise. Define $\mathcal{D}_0 = \{(x, y) : 0 \leq x \leq \sqrt{r(y/x)}\}$. For $X, Y \sim$ Uniform $[0, 1]$, compute the probability of acceptance $\mathrm{P}\left((X, Y) \in \mathcal{D}_0\right)$.
Hint: drawing a picture of $\mathcal{D}_0$ may help.

3. Beta Distribution